Governing Equations

The steady three-dimensional compressible Navier–Stokes equations for an ideal gas are expressed as:

$$\text{Continuity:} \qquad \nabla \cdot (\rho \mathbf{U}) = 0, \tag{1}$$

$$\text{Momentum:} \qquad \nabla \cdot \big(\rho \mathbf{U} \otimes \mathbf{U} + p\mathbf{I} - \boldsymbol{\tau}\big) = 0, \tag{2}$$

$$\text{Total energy:} \qquad \nabla \cdot \big[\mathbf{U}(E + p) - \boldsymbol{\tau} \cdot \mathbf{U} - \mathbf{q}\big] = 0, \tag{3}$$

where the primitive variables are the density $\rho(x, y, z)$, velocity $\mathbf{U} = (u, v, w)$, and pressure $p(x, y, z)$. The temperature, total energy, and enthalpy are given by

$$T = \frac{p}{\rho R}, \qquad E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho|\mathbf{U}|^2, \qquad H = E + p. \tag{4}$$

The viscous stress tensor and heat flux vector are defined as

$$\boldsymbol{\tau} = \mu\left(\nabla\mathbf{U} + (\nabla\mathbf{U})^T - \frac{2}{3}(\nabla\cdot\mathbf{U})\,\mathbf{I}\right), \tag{5}$$

$$\mathbf{q} = -k\,\nabla T, \qquad k = \frac{\mu\, c_p}{\mathrm{Pr}}, \qquad c_p = \frac{\gamma}{\gamma - 1}R. \tag{6}$$

The viscosity μ may be either constant ($\mu = 1/\mathrm{Re}$) or follow the nondimensional Sutherland law:

$$\mu(T) = \mu_0 \frac{T^{3/2}(1 + S)}{T + S}.$$

Boundary Conditions

The computational domain is defined as

$$x \in [0, X_{\max}], \qquad y \in [-Y_{\max}, Y_{\max}], \qquad z \in [-Z_{\max}, Z_{\max}].$$

1. **Inlet (nozzle exit, $x = 0$, $r \le R_e$):**

$$\rho = \rho_e, \qquad u = U_c \cdot \frac{1}{2}\left[1 - \tanh\left(\frac{r - 0.9R_e}{0.08R_e}\right)\right], \qquad v = w = 0, p = p_e. \tag{7}$$

2. **Far-field (outer faces):**

$$\rho = \rho_\infty, \qquad \mathbf{U} = \mathbf{0}, \qquad p = p_\infty. \tag{8}$$

3. **Symmetry plane ($z = 0$):**

$$w = 0, \qquad \frac{\partial \rho}{\partial z} = \frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = \frac{\partial p}{\partial z} = 0. \tag{9}$$

4. **Sponge layer (non-reflecting region):** To reduce wave reflection near boundaries, a spatially weighted penalty term is added:

$$\mathcal{L}_{\text{sponge}} = \int_\Omega \omega(\mathbf{x}) \left[(\rho - \rho_\infty)^2 + |\mathbf{U}|^2 + (p - p_\infty)^2\right] \mathrm{d}\mathbf{x},$$

where $\omega(\mathbf{x}) \in [0, 1]$ increases smoothly toward the domain boundaries.

Physics-Informed Neural Network Loss

The neural network predicts $\{\rho, u, v, w, p\}$ at collocation and boundary points. The total loss combines the PDE residuals and boundary constraints:

$$\mathcal{L}_{\text{total}} = w_{\text{pde}}\,\mathcal{L}_{\text{pde}} + w_{\text{inlet}}\,\mathcal{L}_{\text{inlet}} + w_{\text{far}}\,\mathcal{L}_{\text{far}} + w_{\text{sym}}\,\mathcal{L}_{\text{sym}} + w_{\text{sponge}}\,\mathcal{L}_{\text{sponge}}. \tag{10}$$

Each component is defined as:

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_c}\sum_{j=1}^{N_c}\Big(|\nabla\cdot(\rho\mathbf{U})|^2 + \sum_{i=x,y,z}|\nabla\cdot(\rho U_i\mathbf{U} + p\mathbf{e}_i - \boldsymbol{\tau}_i)|^2 + |\nabla\cdot[\mathbf{U}(E+p) - \boldsymbol{\tau}\cdot\mathbf{U} - \mathbf{q}]|^2\Big), \tag{11}$$

$$\mathcal{L}_{\text{inlet}} = \frac{1}{N_{\text{in}}}\sum\big(|\rho-\rho_e|^2 + |u-u_e|^2 + |v|^2 + |w|^2 + |p-p_e|^2\big), \tag{12}$$

$$\mathcal{L}_{\text{far}} = \frac{1}{N_\infty}\sum\big(|\rho-\rho_\infty|^2 + |\mathbf{U}|^2 + |p-p_\infty|^2\big), \tag{13}$$

$$\mathcal{L}_{\text{sym}} = \frac{1}{N_{\text{sym}}}\sum\big(|w|^2 + |\partial_z\rho|^2 + |\partial_z u|^2 + |\partial_z v|^2 + |\partial_z p|^2\big). \tag{14}$$

Here w_{pde}, w_{inlet}, w_{far}, w_{sym}, and w_{sponge} are tunable weighting coefficients.